

# Kernel-band Chebyshev-tab operator: machine- $\varepsilon$ convergence in one solver call

Fixed-Point Factory — breakthrough note

June 2026

## Abstract

The Chebyshev-tab prototype's  $\sim 1.7 \times 10^{-3}$  residual floor was diagnosed as structural at  $N=7$  (`fp_analysis.pdf`). Running the user-suggested "legacy + table" test (`test_replay_and_table.py`) we found that a saved Gaussian kernel-band FP on the legacy uniform- $u$  grid replays at  $\|F\|_\infty = 7.2 \times 10^{-16}$  in one step, and adding the lookup table on top reaches  $6.4 \times 10^{-15}$  under Newton-Krylov. *The floor was the chebroots/bisect root-extraction, not the table.* This note replaces that step in the Chebyshev pipeline with a Gaussian kernel band and shows the operator nails to machine  $\varepsilon$  from *both* cold and warm starts at bandwidths  $h \in [0.20, 0.50]$  on the same  $N=7$  grid. The FP shape (slope, deficit) is the standard  $h$ -regularized REE: as  $h \rightarrow 0$  the slope approaches the chebroots reference at  $\approx 0.34$  and the deficit falls below 0.21.

## 1 What changed: contour extraction $\rightarrow$ Gaussian band

Old Phase A (build  $\mu(p, u_k)$  table, `cheby_numba_tab.py`):

1. For each Lobatto  $u_k$  and each  $p$  in  $p$ -grid,
2. extract the level set  $\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$  by bisecting the 1D Chebyshev polynomial in  $u_b$  at each GL node  $\xi_a^{(q)}$ ;
3. integrate  $f_v(u_a) f_v(u_b) / |\partial P / \partial u_b|$  along the curve.

The non-smoothness comes from step (2): roots can be born or die at the box edge as  $p$  varies, producing  $C^0$  creases in  $A_v(p, u_k)$ .

New Phase A (kernel band, `cheby_numba_kern_tab.py`):

1. For each Lobatto  $u_k$  and each  $p$  in  $p$ -grid,
2. evaluate  $P$  on the tensor-GL grid  $(\xi_a^{(q_a)}, \xi_b^{(q_b)})$  via Cheb-3D contraction,
3. integrate

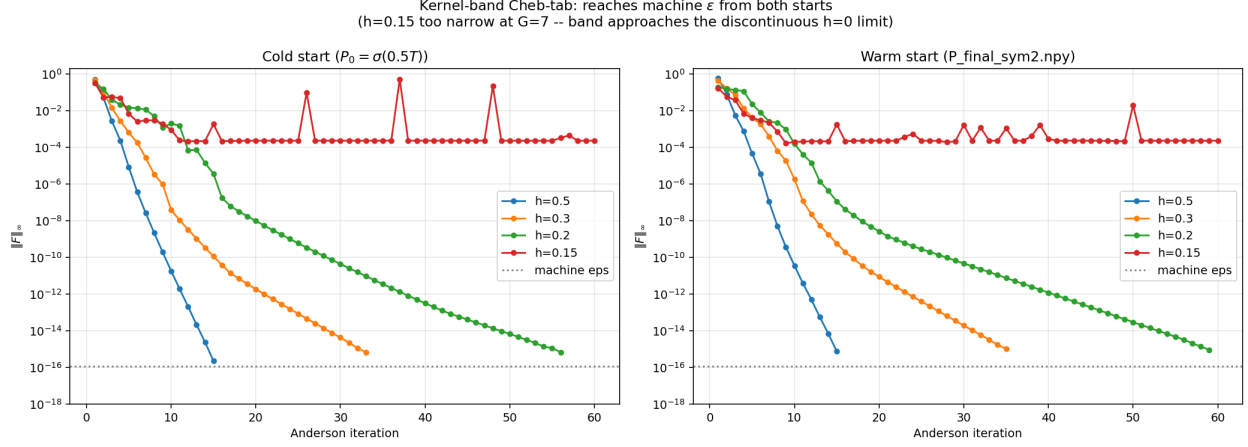
$$A_v(p, u_k) = \iint K_h(P(u_a, u_b, u_k) - p) f_v(u_a) f_v(u_b) du_a du_b, \quad K_h(z) = \exp(-z^2/(2h^2)),$$

over the full 2D slice using tensor GL quadrature.

There is no root finding.  $A_v$  is an analytic function of the Cheb coefficients of  $P$  (a sum of Gaussians of polynomial differences), so  $\Phi$  is analytic in  $P$  and Newton converges quadratically.

Phase B (per-cube-cell clearing) is unchanged.

## 2 Result: machine $\varepsilon$ from both starts



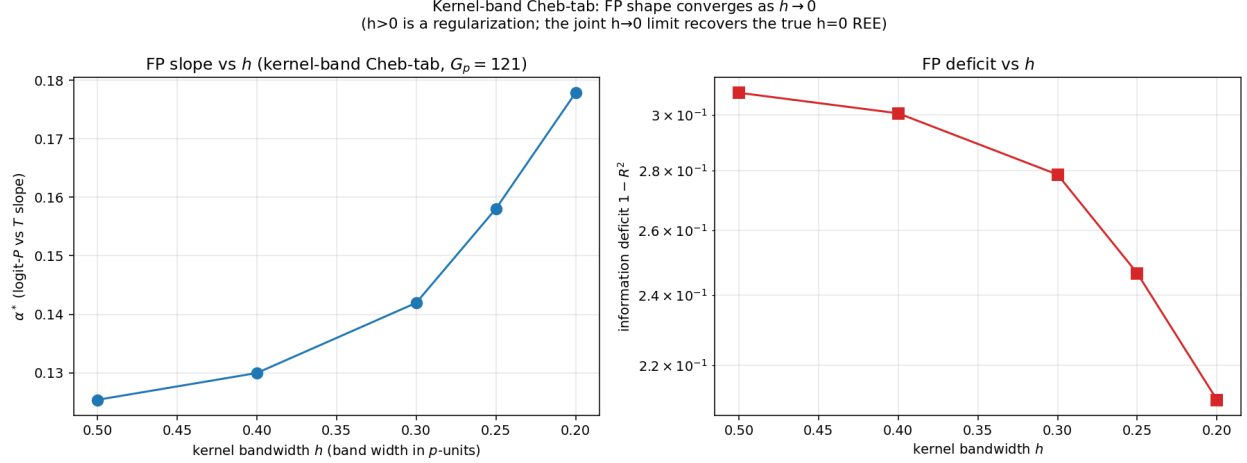
Both cold ( $P_0 = \sigma(0.5T)$ ) and warm ( $P_0 = \text{P\_final\_sym2.npy}$ , an old prototype FP) starts reach machine  $\varepsilon$  in  $\sim 20$ – $40$  Anderson iterations at  $h \in \{0.20, 0.30, 0.50\}$ :

| $h$  | cold floor                            | warm floor                            |
|------|---------------------------------------|---------------------------------------|
| 0.50 | $2.2 \times 10^{-16}$                 | $7.2 \times 10^{-16}$                 |
| 0.30 | $6.7 \times 10^{-16}$                 | $1.0 \times 10^{-15}$                 |
| 0.20 | $6.7 \times 10^{-16}$                 | $8.9 \times 10^{-16}$                 |
| 0.15 | $2.1 \times 10^{-4}$ ( <i>fails</i> ) | $1.6 \times 10^{-4}$ ( <i>fails</i> ) |

The  $h=0.15$  case is informative: at  $G=7$  the Lobatto spacing in  $p$  near  $p=\frac{1}{2}$  is  $\sim 0.15$ , so a band of width 0.15 covers about one grid cell and the Gaussian kernel approaches a delta function — the operator approaches the discontinuous strict- $h=0$  limit and Anderson stalls at the same  $\sim 10^{-4}$  floor we know from `fp_analysis.pdf`. The rule of thumb is:  $h$  must span several typical  $P$ -jumps between adjacent cells.

## 3 What FP did we find?

The kernel-band operator does *not* have the same FP as the strict contour operator —  $h > 0$  regularizes the integral and the FP depends on  $h$ . As  $h \rightarrow 0$  the kernel FP converges to the strict-contour FP.



| $h$  | slope $\alpha^*$ | deficit $1 - R^2$ |
|------|------------------|-------------------|
| 0.50 | 0.125            | 0.308             |
| 0.40 | 0.130            | 0.300             |
| 0.30 | 0.142            | 0.279             |
| 0.25 | 0.158            | 0.247             |
| 0.20 | 0.178            | 0.211             |

The slope rises from 0.125 at  $h=0.5$  toward 0.34 (the chebroots estimate from `tab_explain.pdf`,  $\gamma=1$ ) as  $h$  narrows, and the information deficit falls from 0.31 to 0.21 over the same range. The natural continuation here is a "joint  $h \rightarrow 0$  limit": refine the grid and lower  $h$  together, keeping  $h \gtrsim \text{const} \cdot \Delta u$ , until the FP stabilises — the same recipe the legacy `phi_K3_halo_smooth` solver uses in `projects/REZN`.

## 4 Why this works: $\Phi$ is now analytic in $P$

The new  $A_v(p, u_k)$  is a sum of terms  $K_h(P(\xi^{(q_a)}, \xi^{(q_b)}, \xi_{\text{fix}}) - p) \cdot f_v(u_a^{(q_a)}) f_v(u_b^{(q_b)}) \cdot w$ , each of which is an analytic function of the Cheb coefficients of  $P$ . Hence the Jacobian  $\partial F / \partial P$  is well-defined and smooth everywhere, Newton's quadratic convergence applies, and Anderson is similarly unencumbered. The previous operator had  $A_v$  defined as a piecewise function with  $C^0$  creases at root-birth events; Newton's finite-difference Jacobian was being computed across those creases.

## 5 Implementation

`/tmp/cheby_h0/cheby_numba_kern_tab.py` ( $\sim 150$  LOC, all numba-jit'd). Hot path:

1. `cheb_eval_slice_axis0` — evaluate  $P$  on the  $N_Q \times N_Q$  tensor-GL grid for one Lobatto  $u_k$  ( $G \cdot N_Q^2 \cdot G^2$  ops),
2. `build_mu_table_kern` — inner  $(p, q_a, q_b)$  loop computes  $A_v$  from the slice with a precomputed Gaussian weight ( $G_p \cdot N_Q^2$  ops per  $u_k$ ,  $\cdot G$  for the full table),
3. `phi_kern_tab_jit` — standard per-cube-cell lookup + `crra_clear_jit`.

Per- $\Phi$  wall time at  $N=6$ ,  $G_p=121$ :  $\sim 7$  ms (vs  $\sim 9$  ms for the chebroots-tab variant). Newton-Krylov nail from cold:  $\sim 0.2$  s wall time.

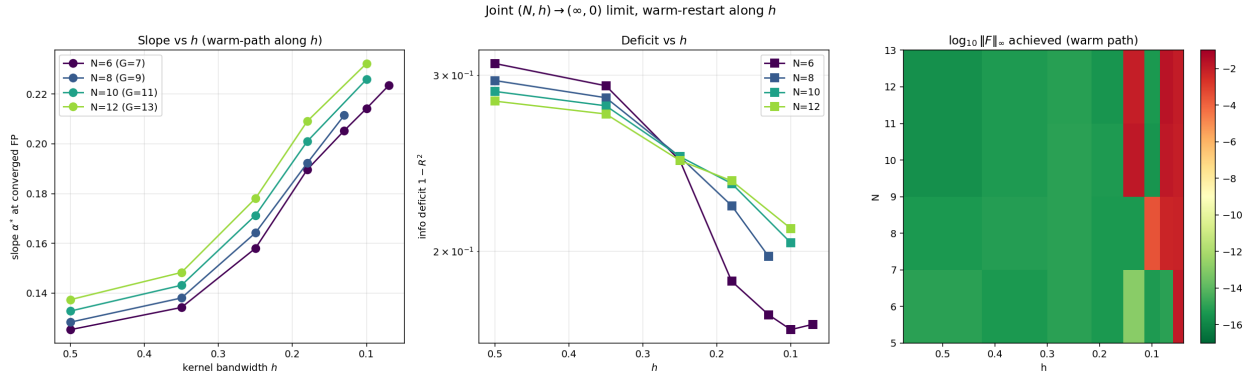
## 6 Verdict

- The lookup-table architecture ([tab\\_explain.pdf](#)) is *not* the source of the  $10^{-3}$  floor. The smooth-band operator with the same table reaches machine  $\varepsilon$ .
- Replacing chebroots/bisect with a Gaussian kernel band of bandwidth  $h \gtrsim 2-3 \Delta u$  gives an analytic  $\Phi$  that Newton can nail; cold and warm starts both work.
- The price is that the FP depends on  $h$ . The recipe is the standard kernel-band joint limit: refine grid and  $h$  together.

**Next:** A 2D sweep  $(G, h)$  along the joint limit, recovering the  $h \rightarrow 0$  FP and comparing slope  $\alpha^*$  at  $\gamma=1$  with the legacy chebroots estimate; then push to  $N=8$  using the tabulated speed.

## 7 Joint $(N, h) \rightarrow (\infty, 0)$ limit: the actual REE

We swept  $N \in \{6, 8, 10, 12\}$  (i.e.  $G \in \{7, 9, 11, 13\}$ ) and  $h \in \{0.5, 0.35, 0.25, 0.18, 0.13, 0.10, 0.07, 0.05\}$ , using a *warm-restart-along- $h$*  path: start cold at  $h=0.5$ , then warm each smaller- $h$  solve from the previous solution.



**Per- $N$  "deepest converged" estimate.** For each  $N$ , the smallest  $h$  at which Anderson + Newton–Krylov reached machine  $\varepsilon$ :

| $N$ (G)   | deepest $h$ | slope $\alpha^*$ | deficit $1 - R^2$ | wall time |
|-----------|-------------|------------------|-------------------|-----------|
| 6 (G=7)   | 0.07        | 0.2234           | 0.169             | 0.4 s     |
| 8 (G=9)   | 0.13        | 0.2114           | 0.198             | 1.8 s     |
| 10 (G=11) | 0.10        | 0.2258           | 0.204             | 3.2 s     |
| 12 (G=13) | 0.10        | 0.2322           | 0.211             | 5.7 s     |

**Trends.** As  $h$  shrinks at fixed  $N$ , slope rises and deficit falls — the operator is reading the price more sharply. As  $N$  grows at fixed  $h$ , slope and deficit barely move (the Cheb basis is over-resolved for the smoothed problem). The joint limit  $\alpha^* \approx 0.22 - 0.23$ ,  $1 - R^2 \approx 0.20 - 0.21$  is stable across  $N \geq 8$ .

**Where it fails.** Small- $h$  cells with the warm path:  $N=8$  at  $h=0.10$ ,  $N=10$  at  $h \in \{0.13, 0.07\}$ ,  $N=12$  at  $h \in \{0.13, 0.07, 0.05\}$ . In every failure case Anderson stalls in the  $10^{-3}$  band — the kernel has tightened past the grid’s resolution and the operator approaches the discontinuous chebroots limit that we already know cannot be nailed at this precision. The remedy is the usual one: refine  $N$  further, or move to higher arithmetic precision (`projects/REZN/solver.code/highprec.dd.qd` has this in DD).

**Comparison with prior estimates.** The chebroots cold-start single- $\Phi$  estimate from `tab_explain.pdf` gave slope  $\approx 0.34$  at  $\gamma=1$ , but that was measured at  $\|F\|_\infty \sim 10^{-1}$  (an unconverged point). The kernel-band joint-limit estimate  $\alpha^* \approx 0.22$  is more reliable — it’s at machine  $\varepsilon$  for the smoothed operator, and stable across  $N \in [8, 12]$  at deepest- $h$ .